

Chapter 1: Algebraic Background

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Exercise 1.1

Domain $\mathbb{Z}$ is not a field.

Exercise 1.2

- a) irreducible
- b) $(x + 13)(x - 13)$
- c) $(x + 1)(x^2 + 1)$
- d) Irreducibility can be obtained by computing the cases of $x \in \{-3, -2, -1, 0, 1\}$ and checking monotonicity.

Exercise 1.3

$$x^2 - 1 = (x + 1)(x - 1)$$

Exercise 1.4

Yes.

In fact, for any field F and $f(x) \in F[x]$. We'll prove $\exists K/F$, s.t. f have multiple roots in $K \iff f, f'$ are not relatively prime: Let α be a root of f , $f(x) = (x - \alpha)g(x)$. Now $f'(x) = g(x) + (x - \alpha)g'(x)$. Substituting $x = \alpha$, we'll see that $f'(\alpha) = 0 \iff g(\alpha) = 0 \iff f(x)$ has multiple roots.

Exercise 1.5

- i. $x^4 + 1$
- ii. $x^4 - 2x^2 + 9$
- iii. $x^3 - 6x^2 + 12x - 9$

Exercise 1.6

- a) 2
- b) 1
- c) 4
- d) 3
- e) infinite

Exercise 1.7

It is an algebraic extension for sure since its generators are all algebraic elements.

However, it's not finite since the minimal polynomial of each of the generators can be arbitrarily high.

Exercise 1.8

- a) $s_1^2 - 2s_2$
- b) $s_1^3 - 3s_1s_2$

- c) It's not symmetric.
- d) It's not symmetric.

Exercise 1.9

- i. We can prove Δ is antisymmetric by comparing the definition of even permutations and components of Δ .
- ii. First, we'll prove that for $f(x_1, x_2, \dots, x_n) \in \mathbb{Z}[x_1, x_2, \dots, x_n]$ which is antisymmetric, $\Delta \mid f(x_1, x_2, \dots, x_n)$: To prove this, we only have to prove that $\exists i, j \in [1, n] \cap \mathbb{Z}$, let's say $i = 1, j = 2, (x_i - x_j) \mid f(x_1, x_2, \dots, x_n)$. Notice $f(x_1, x_2, \dots, x_n) \in (\mathbb{Z}[x_3, x_4, \dots, x_n])[x_1, x_2]$, we can have

$$f(x_1, x_2, \dots, x_n) = \sum_{i, j \in [0, n] \cap \mathbb{Z}} a_{ij} x_1^i x_2^j$$

where $a_{ij} \in \mathbb{Z}[x_3, x_4, \dots, x_n]$. And because $f(x_1, x_2, \dots, x_n) = -f(x_2, x_1, \dots, x_n)$, $a_{ij} = -a_{ji}$, indicating $(x_1 - x_2) \mid f(x_1, x_2, \dots, x_n)$. Thus, $\Delta \mid f(x_1, x_2, \dots, x_n)$.

Now, following from the hint, we consider f/Δ . It is a symmetric polynomial, so it was consist of elementary symmetric polynomials.

Exercise 1.10 [I'm not quite sure about (b) and (d)]

a) 42

b) $\begin{bmatrix} 1 & 3 & 5 \\ 2 & -4 & \\ 7 & 2 & 9 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 5 \\ & 10 & -10 \\ & 11 & 14 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 5 \\ & 1 & 24 \\ & 10 & -10 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 5 \\ & 1 & 24 \\ & & -250 \end{bmatrix}$, the answer is 250.

c) infinite

d) $\begin{bmatrix} 41 & 32 & -999 \\ & 16 & 3 \\ & 2 & 111 \end{bmatrix} \rightarrow \begin{bmatrix} 41 & 50 & 0 \\ & 16 & 3 \\ & 2 & 111 \end{bmatrix} \rightarrow \begin{bmatrix} 41 & 2 & -9 \\ & 16 & 3 \\ & 2 & 111 \end{bmatrix} \rightarrow \begin{bmatrix} 41 & 2 & -9 \\ & 2 & 111 \\ & & -885 \end{bmatrix}$, the answer is $412885 = 72,570$.

e) infinite

Exercise 1.11

- i. The proof can be obtained by checking the definitions.
- ii. The proof can be obtained by checking the definitions.
- iii. I suppose it can't because we no longer have $1m = m$, which means that the values of multiplication km are no longer distributed to the vectors in M .

Exercise 1.12

They are $n\mathbb{Z}$.

Exercise 1.13

No. For example, $\mathbb{Z}$ has infinite numbers of submodules $n\mathbb{Z}$.
Such condition can be obtained by having R as a finite ring.

Exercise 1.14

Exercise 1.15